

University of Brasília

Institute of Exact Sciences
Department of Computer Science

TransferKAN: Ballistic Wound Classification via Transfer Learning and Kolmogorov-Arnold Networks

Arthur Silva Carneiro

Thesis submitted as a partial requirement
for completion of the Bachelor of Science in Computer Science

Advisor
Prof. Dr. Pedro Garcia Freitas

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Dedication

This work is dedicated to the individuals whose presence, sacrifice, and unwavering support made my academic journey possible. I dedicate this work to all of you, with sincere appreciation and respect.

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I would like to express my sincere gratitude to the University of Brasília for granting me the opportunity to pursue my dream degree in Computer Engineering. It has been a privilege to study and develop academically at such a distinguished institution.

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Disclaimer

During the preparation of this work the author used ChatGPT™, Claude™, and Grammarly™ to review English usage and grammatical correction. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Resumo

A análise de ferimentos por arma de fogo (Entrada e Saída) e a estimativa da distância do disparo (MLSD - Medical-Legal Shooting Distance) são tarefas periciais desafiadoras, geralmente demandando especialistas e dependendo de condições sub-ideais de captura de imagens em cenas de crime. O advento do Aprendizado Profundo (Deep Learning) trouxe novas oportunidades para automatizar e auxiliar essas tarefas, mas ainda enfrenta dificuldades com bases de dados limitadas e desbalanceadas. Este trabalho investiga o uso de uma arquitetura híbrida denominada TransferKAN para a classificação de ferimentos balísticos em imagens reais de cenas de crime da base de dados FDCPUnBGunshotDB. A arquitetura proposta utiliza uma rede convolucional pré-treinada (ResNet152) para extração de características espaciais robustas, substituindo o classificador denso tradicional por uma Kolmogorov-Arnold Network (KAN), capaz de modelar relações não-lineares complexas de forma mais eficiente.

A abordagem foi avaliada sob um esquema de validação cruzada k-fold rigoroso, prevenindo o vazamento de dados, e acompanhada de técnicas de Data Augmentation avançadas para mitigar o desbalanceamento inerente às amostras forenses. O desempenho do modelo TransferKAN foi comparado com resultados da literatura (artigo de 2024), demonstrando superioridade técnica.

Para a tarefa de classificação do tipo de ferimento (Entrada vs. Saída), a TransferKAN com validação k-fold obteve 86,0% de Acurácia e uma AUC ponderada de 0,908, com F1-Score ponderado de 0,862. Na classificação da distância do disparo (MLSD), a TransferKAN apresentou uma melhoria em precisão ponderada, recall ponderado e F1-score balanceado, obtendo 87,3% de Acurácia. Esses resultados evidenciam que a TransferKAN obteve a melhor performance e os melhores resultados gerais em métricas integradas, destacando o potencial da utilização das redes KAN como classificadores no auxílio de patologia forense em cenários reais.

Palavras-chave: Ferimentos Balísticos, Aprendizado por Transferência, Redes de Kolmogorov-Arnold (KAN), Visão Computacional

Abstract

The analysis of gunshot wounds (Entry and Exit) and the estimation of the firing distance (MLSD - Medical-Legal Shooting Distance) are challenging forensic tasks. They typically demand highly specialized experts and depend on sub-optimal image capture conditions at crime scenes. The advent of Deep Learning has brought new opportunities to automate and assist in these tasks, yet it still faces difficulties with limited and imbalanced forensic datasets. This work investigates the use of a hybrid architecture called TransferKAN for the classification of ballistic wounds on real crime scene images from the FDCPUnBGunshotDB dataset. The proposed architecture employs a pre-trained convolutional neural network (ResNet152) to extract robust spatial features, replacing the traditional dense classifier with a Kolmogorov-Arnold Network (KAN), which is capable of modeling complex non-linear relationships more efficiently.

The approach was evaluated under a rigorous k-fold cross-validation scheme to prevent data leakage, coupled with advanced Data Augmentation techniques to mitigate the inherent class imbalance of forensic samples. The performance of the TransferKAN model was compared with results from literature (a 2024 paper baseline), demonstrating technical superiority.

For the wound type classification task (Entry vs. Exit), the TransferKAN model with k-fold validation achieved an Accuracy of 86.0% and a weighted AUC of 0.908, with a weighted F1-Score of 0.862. In the firing distance classification (MLSD), the TransferKAN showed improvements in weighted precision, weighted recall, and balanced F1-score, reaching 87.3% Accuracy. These findings demonstrate that our TransferKAN approach with k-fold validation achieved the best performance and overall results, highlighting the potential of using KAN networks as classifiers to assist forensic pathology in real-world scenarios.

Keywords: Ballistic Wounds, Transfer Learning, Kolmogorov-Arnold Networks (KAN), Computer Vision

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Chapter 1

Introduction

1.1 Problem Statement

Firearms correspond to one of the most frequently used instruments worldwide for committing homicides [1]. In forensic fieldwork, encountering gunshot wounds during external body examinations is common, whether at crime scenes or during autopsies. During these examinations, forensic experts must precisely determine the nature of the wound. The primary challenges involve distinguishing between entry and exit wounds and classifying the entry wounds according to the distance from which the shot was fired, an assessment commonly referred to as the Medical-Legal Shooting Distance (MLSD).

Under ideal conditions, certain morphological characteristics—such as shape, size, presence of an abrasion ring, and margins—allow for a relatively easy differentiation between entry and exit wounds. However, in reality, these characteristics are not always clearly visible; they may be partially absent, overlapping, or drastically altered by putrefaction. Other confounding factors, including the firearm caliber, ammunition type, shot angle, intermediate targets, and the impacted body region, further complicate accurate classification, even for experienced forensic pathologists.

The application of Artificial Intelligence (AI) and Deep Learning (DL) in computer vision tasks has recently revolutionized medical diagnostics, offering robust and automated tools for complex image classification. Despite these remarkable advancements in clinical settings, the translation of such technologies into the forensic domain remains extremely limited. One of the principal challenges lies in acquiring comprehensive datasets with high-quality, scientifically sound data. Forensic images are usually taken under various environmental conditions, with a range of photographic equipment, and within the chaotic, routine circumstances of crime scenes, making standardization practically impossible and creating highly imbalanced datasets.

Existing literature in the forensic domain, such as recent classification models trained on human autopsy databases [2, 3], relies heavily on traditional Convolutional Neural Networks (CNNs). These architectures universally employ conventional Multi-Layer Perceptrons (MLPs) comprising dense linear layers for their final classification stages. While these architectures perform adequately under generalized circumstances, they inherently struggle to efficiently map the highly non-linear, complex feature boundaries present in noisy, imbalanced, and unstandardized forensic wound images. There is a critical gap and pressing need for a more advanced, efficient classifier that can extract and separate abstract non-linear patterns without requiring exponentially larger datasets or massive parameter counts.

1.2 Research Objectives

The primary objective of this study is to develop, implement, and evaluate a novel hybrid deep learning architecture, named **TransferKAN**, for the automated classification of human gunshot wounds using real crime scene images. This work focuses meticulously on two specific forensic tasks: classifying wound types (Entry vs. Exit) and determining the Medical-Legal Shooting Distance (MLSD) as Contact, Close range, or Distant.

To achieve this overarching goal, the following specific objectives were established:

- To implement a Transfer Learning framework utilizing a pre-trained ResNet152 backbone, capitalizing on its proven capacity for high-level spatial feature extraction from complex images.
- To design and integrate a Kolmogorov-Arnold Network (KAN) as the primary decision classifier, replacing traditional dense MLP layers, in order to better capture and parameterize complex non-linear relationships through learnable splines.
- To implement a rigorous, case-based k-fold cross-validation scheme to evaluate the model's performance reliably, eliminating any possibility of semantic data leakage that typically inflates metrics in forensic AI research.
- To rigorously compare the performance metrics (Accuracy, Precision, Recall, F1-Score, AUC, and Specificity) of the proposed TransferKAN architecture against the baseline performance of state-of-the-art CNNs, specifically the ResNet152 baseline model reported in the benchmark study of Lira et al. [2].

1.2.1 Research Hypothesis

We hypothesize that replacing the standard Multi-Layer Perceptron (MLP) classification head of a state-of-the-art deep convolutional network (ResNet152) with a Kolmogorov-Arnold Network (KAN) will result in a superior, highly robust hybrid architecture (TransferKAN). By learning activation functions on the network's edges rather than relying on fixed node activations, the TransferKAN architecture will better model the non-linear boundaries of imbalanced forensic image classes. Consequently, this approach will yield higher discriminative performance and reliability—particularly evidenced by the Area Under the Curve (AUC)—in both wound type and MLSD classification compared to the traditional ResNet152 baseline.

1.3 Contributions

The main contributions of this work can be summarized as follows:

- The pioneering introduction of the **TransferKAN architecture** into the forensic pathology domain. This hybrid model successfully leverages the deep feature-extraction prowess of ResNet152 alongside the parameter-efficient, dynamic non-linear classification capabilities of Kolmogorov-Arnold Networks.
- The empirical demonstration that the TransferKAN approach, coupled with robust data augmentation and strict k-fold cross-validation, decisively outperforms the traditional CNN baselines utilized in prior forensic literature, particularly showing vast improvements in robustness and minority class detection for Entry/Exit and MLSD classifications on a real-world crime scene dataset.
- The provision of a rigorous methodological pipeline that actively mitigates data leakage issues through case-based data splitting, offering a much more realistic evaluation of the model's true generalization capabilities in authentic forensic investigative scenarios.

1.4 Organization of This Document

The remainder of this dissertation is organized as follows. Chapter 2 provides the essential background on the forensic pathology of gunshot wounds and introduces the theoretical underpinnings of Deep Learning and Kolmogorov-Arnold Networks (KAN). Chapter 3 critically reviews the related literature on AI applied to forensic wound classification, identifying current gaps. Chapter 4 introduces the proposed solution, detailing the TransferKAN architecture. Chapter 5 describes the proposed methodology, including the dataset,

data augmentation, and cross-validation procedures. Chapter 6 presents a comprehensive comparative analysis of the results between TransferKAN and the established baselines. Finally, Chapter 7 concludes the thesis, summarizing the findings and suggesting potential avenues for future research.

Chapter 2

Background

This chapter provides the essential theoretical foundation required to understand the interdisciplinary nature of this work, bridging forensic pathology and advanced artificial intelligence. It details the fundamental concepts of gunshot wound classification, the mechanisms of deep learning for computer vision, and the mathematical theory underlying Kolmogorov-Arnold Networks (KAN).

2.1 Forensic Pathology of Gunshot Wounds

In forensic medicine, the accurate analysis of gunshot wounds is crucial for reconstructing the events of a crime scene. When a projectile strikes a human body, it produces a predictable set of morphological alterations that vary significantly depending on the firearm, the ammunition, and the distance from which it was fired [4, 5].

2.1.1 Entry versus Exit Wounds

The most fundamental classification tasks involve distinguishing an entry wound from an exit wound. An entry wound is typically characterized by an *abrasion ring* (a marginal rim of scraped skin), tissue inversion, and a generally regular, circular, or oval shape, depending on the angle of incidence. The physical properties of the skin being pushed inward by the bullet cause these specific signs. Conversely, an exit wound usually presents with everted (outward-turning) tissue, irregular or stellate (star-like) shapes, and the absence of an abrasion ring or gunshot residue, as the projectile pushes the tissue outward from within.

However, intermediate targets, skeletal deflections, and postmortem changes such as putrefaction can obscure these characteristics, making visual classification a highly complex task heavily reliant on expert forensic experience.

2.1.2 Medical-Legal Shooting Distance (MLSD)

Beyond distinguishing entry from exit, classifying the distance of the shot—the Medical-Legal Shooting Distance (MLSD)—is vital for legal investigations. MLSD is categorized based on macroscopic traces left by the combustion of gunpowder and primer components on the skin [4]:

- **Contact:** The muzzle of the firearm is pressed against the skin. These wounds often display a "muzzle imprint" and severe tissue lacerations due to expanding gases entering the subcutaneous layers.
- **Close Range:** Fired from a short distance (typically less than 60 cm), these wounds exhibit *tattooing* or *stippling* (unburned powder grains embedded in the skin) and soot deposition around the entry hole.
- **Distant:** Fired from a distance where gunpowder residues no longer reach the skin. The wound primarily shows the abrasion ring without surrounding soot or stippling.

2.2 Deep Learning for Computer Vision

Deep Learning (DL), a subset of machine learning, has revolutionized the field of Computer Vision by enabling computational models to automatically discover and extract the representations needed for feature detection from raw data.

2.2.1 Convolutional Neural Networks (CNNs) and ResNet

Convolutional Neural Networks (CNNs) are the standard architecture for image classification. They utilize convolutional layers to extract hierarchical spatial features—from simple edges in the lower layers to complex semantic shapes in the deeper layers.

A persistent issue in training deep CNNs is the vanishing gradient problem, where gradients become too small to update the weights effectively in early layers. The Residual Network (ResNet) architecture [6] solved this by introducing "skip connections" (or residual blocks), which allow the gradient to bypass certain layers. ResNet152, a variant with 152 layers, represents a pinnacle of deep feature extraction, widely used as a robust backbone in transfer learning applications for medical and forensic imaging, typically initialized with weights pre-trained on the large-scale ImageNet database [7].

Traditionally, the deep spatial features extracted by architectures like ResNet are flattened and passed through a Multi-Layer Perceptron (MLP)—composed of dense, fully connected linear layers followed by fixed non-linear activation functions (e.g., ReLU)—to produce the final class probabilities.

2.3 Kolmogorov-Arnold Networks (KAN)

While MLPs have been the ubiquitous standard for the classification head in deep learning, they are fundamentally based on the Universal Approximation Theorem. Recently, a novel neural network architecture known as Kolmogorov-Arnold Networks (KAN) has been proposed [8], inspired by the Kolmogorov-Arnold representation theorem.

2.3.1 Theoretical Foundation

The Kolmogorov-Arnold representation theorem states that any multivariate continuous function can be represented as a finite composition of continuous functions of a single variable and the operation of addition.

In a traditional MLP, the network learns fixed weights (linear transformations) on the edges, and applies a fixed, predefined non-linear activation function (like ReLU or Sigmoid) at the nodes. In stark contrast, a KAN flips this paradigm: it places *learnable* non-linear activation functions on the *edges* (connections) and simply sums them at the nodes.

2.3.2 B-Splines and Parameter Efficiency

To make the edge functions learnable, KANs typically parameterize them using B-splines (Basis splines) [8]. A B-spline is a piecewise polynomial function that can flexibly adapt its shape to model highly complex, non-linear relationships.

Because every weight parameter in an MLP is replaced by a highly flexible spline function in a KAN, KANs generally require far fewer parameters to achieve the same or better mapping of complex decision boundaries. This makes them highly parameter-efficient and less prone to overfitting on imbalanced or relatively small datasets—a critical advantage for forensic computer vision, where obtaining thousands of perfectly standardized gunshot wound images is practically impossible.

Chapter 3

Related Work

This chapter reviews the current literature concerning the application of artificial intelligence to forensic pathology, specifically focusing on the classification of gunshot wounds. It critically analyzes the state-of-the-art benchmark established by recent studies and identifies the architectural gaps that the proposed TransferKAN model aims to address.

3.1 AI in Forensic Pathology

The integration of artificial intelligence into forensic sciences has been historically slower compared to clinical medicine. Initial studies in the domain predominantly utilized animal carcasses, such as pigs, to simulate human skin properties when analyzing gunshot wounds [9]. While these controlled laboratory environments allowed for high classification accuracies, they lacked the extreme variability, contamination, and complex morphological alterations encountered in actual human crime scenes. Subsequent work moved towards human samples: Cheng et al. [3] trained classifiers on autopsy photographs of human gunshot wounds, but these images were obtained under the relatively controlled conditions of necropsy examinations and did not address the MLSD task.

3.2 The 2024 Benchmark Study

A significant breakthrough in the field was recently achieved by Lira et al. in their paper *Deep learning-based human gunshot wounds classification* [2]. This study was pioneering because it utilized a substantial, real-world forensic dataset—the FDCPUnBGunshotDB—comprising 2,551 digital photographs taken directly from crime scenes and autopsies by the Federal District Civil Police in Brazil.

The authors systematically evaluated 59 different deep learning architectures (including various iterations of EfficientNet, MobileNet, ViT, and ResNet) to perform two pri-

mary tasks: distinguishing between entry and exit wounds, and classifying the Medical-Legal Shooting Distance (MLSD).

3.2.1 Baseline Methodologies and Results

In their methodology, Lira et al. employed extensive data augmentation to combat the inherent imbalance of the dataset (e.g., 1,883 entry wounds versus only 668 exit wounds). They utilized a hold-out cross-validation approach and found that the **ResNet152** architecture demonstrated the superior overall performance across both tasks.

- For the **Entry vs. Exit** binary classification, the ResNet152 achieved an accuracy of 86.90% and an Area Under the Curve (AUC) of 82.09%.
- For the **MLSD** multiclass categorization, the same architecture achieved an impressive accuracy of 92.48% and an AUC of up to 94.36%.

3.3 Identifying the Research Gap

While the 2024 study established a robust benchmark and proved the viability of computer vision in real-world forensic investigations, a critical architectural limitation persists.

The state-of-the-art ResNet152 model, like all traditional CNNs evaluated in the benchmark, relies on a standard Multi-Layer Perceptron (MLP) for its final classification head. MLPs consist of dense linear layers followed by fixed, non-linear activation nodes. In scenarios involving highly imbalanced forensic datasets where visual features are obscured by noise, blood, and putrefaction, the rigid node-based activations of MLPs require massive amounts of parameters to successfully map highly complex, non-linear decision boundaries.

The authors of the 2024 study noted that sample imbalance still negatively affected metrics such as precision and recall for minority classes. Therefore, there is a clear and pressing need for a lightweight yet highly expressive classification head that can better probabilistically discriminate these complex non-linear boundaries without exponentially increasing the dataset size. This gap forms the direct motivation for introducing Kolmogorov-Arnold Networks (KAN) as the classification head in this thesis.

Chapter 4

Proposed Solution

To address the limitations of traditional CNN architectures in classifying highly imbalanced and non-linear forensic gunshot wound datasets, this thesis introduces a novel hybrid architecture: the **TransferKAN**.

4.1 The TransferKAN Architecture

The core philosophy behind TransferKAN is to merge the proven spatial feature extraction capabilities of state-of-the-art Convolutional Neural Networks with the highly expressive and parameter-efficient classification potential of Kolmogorov-Arnold Networks (KAN).

Traditional architectures, such as the ResNet152 [6] used in the 2024 benchmark [2], process an image through a series of convolutional blocks to generate a high-dimensional feature vector. This vector is then typically fed into a Multi-Layer Perceptron (MLP) classification head. While the ResNet backbone is exceptionally adept at recognizing visual patterns (edges, textures, shapes), the MLP head often struggles to map these features into discrete classes when the decision boundaries are highly complex, noisy, and non-linear—as is inherently the case with forensic imagery of gunshot wounds.

4.1.1 Architectural Components

The TransferKAN architecture consists of three distinct sequential stages:

1. **Spatial Feature Extractor (ResNet152 Backbone):** We utilize a ResNet152 model [6] pre-trained on ImageNet [7]. The final fully connected dense layer (the original MLP classifier) is programmatically removed. Consequently, for a standard input image, the backbone outputs a flattened feature vector of size 2048. This leverages the paradigm of Transfer Learning, avoiding the need to train a deep feature extractor from scratch on a limited forensic dataset.

2. **Regularization (Dropout):** To bridge the backbone and the new classifier while aggressively preventing overfitting, a Dropout layer with a probability of $p = 0.5$ is introduced. This forces the network to learn redundant and robust representations rather than memorizing the training samples.

3. **Kolmogorov-Arnold Network (KAN) Classifier:** The core innovation lies in appending a KAN module [8] directly to the feature vector. Instead of a dense layer matrix multiplication with fixed node activations, the KAN maps the 2048-dimensional vector into the desired number of output classes (e.g., 2 for Entry/Exit or 3 for MLSD) through an intermediate hidden layer of 128 dimensions. The connections between these layers are modeled as learnable B-splines.

The complete data flow of the proposed architecture is summarized in Figure 4.1.

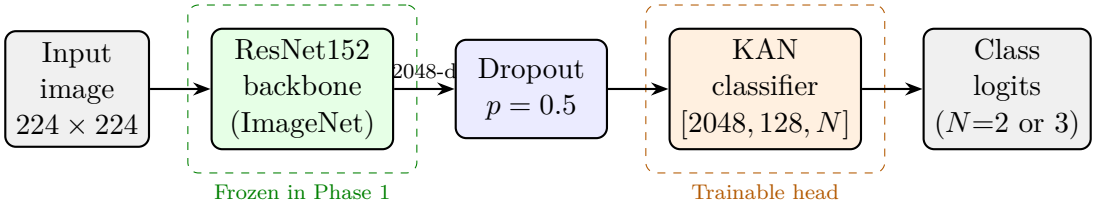


Figure 4.1: The proposed TransferKAN architecture. A ResNet152 backbone pre-trained on ImageNet extracts a 2048-dimensional feature vector, which is regularized by a dropout layer ($p = 0.5$) and mapped to the output classes by a Kolmogorov-Arnold Network classifier with a $[2048, 128, N]$ structure ($N = 2$ for Entry/Exit, $N = 3$ for MLSD).

4.2 Justification of the Innovation

The transition from an MLP classification head to a KAN classifier is not merely a structural substitution; it represents a fundamental shift in how the network models the final decision boundary.

By placing the non-linear activation functions on the edges (splines), the KAN dynamically molds its activation shapes to best fit the data during training. This provides two significant advantages in the forensic context:

- **Superior Discrimination in Imbalanced Data:** Splines allow the KAN to create highly intricate, non-linear boundaries around the minority class clusters (such as Exit wounds or Contact shots) without collapsing into the majority class, a common failure point for rigid MLPs.

- **Parameter Efficiency:** The KAN requires significantly fewer parameters to achieve a highly complex mathematical mapping compared to an MLP, reducing the risk of over-parameterizing and overfitting the limited FDCPUnBGunshotDB dataset.

Through this strategic architectural design, TransferKAN aims to serve as an optimized, probabilistic discriminator tailored specifically to the nuances of forensic pathology.

Chapter 5

Methodology

This chapter details the data acquisition, the architectural implementation of the Transfer-KAN model, and the rigorous experimental setup used to train and validate the proposed architecture. The methodology is meticulously designed to ensure strict data separation, completely preventing the semantic data leakage that often plagues medical image classification tasks.

5.1 Dataset Description

The dataset utilized in this study is the FDCPUnBGunshotDB [2], a curated collection of real crime scene and autopsy photographs obtained by the Federal District Civil Police (PCDF) in Brazil. Unlike datasets composed of staged laboratory environments or animal models, this dataset authentically reflects the severe class imbalances, lighting variations, and morphological distortions naturally occurring in forensic fieldwork.

For the primary **Wound Type Classification** task (Entry vs. Exit), the dataset contains a total of 2,551 images. This is severely imbalanced, comprising:

- **Entry Wounds (Label 0):** 1,883 images
- **Exit Wounds (Label 1):** 668 images

For the secondary **Medical-Legal Shooting Distance (MLSD) Classification** task, only the 1,883 entry wound images are considered, as exit wounds do not display distance characteristics. These are categorized into three distinct, highly skewed classes:

- **Distant (Label 2):** 1,609 images (85.45%)
- **Close range (Label 1):** 232 images (12.32%)
- **Contact (Label 0):** 42 images (2.23%)

Beyond the class imbalance, the dataset reflects the demographic profile typical of firearm-related fatalities: the victims are predominantly young adult males, with a mean age of approximately 27 years and a pronounced right skew towards individuals in their twenties, as shown in Figure 5.1.

5.2 TransferKAN Implementation Details

The core of our proposed solution is the **TransferKAN** architecture, implemented using PyTorch [10].

5.2.1 Backbone and Regularization

We employed a pre-trained ResNet152 [6] from TorchVision as the backbone for spatial feature extraction, leveraging its pre-existing weights to identify complex low-level visual patterns without requiring a massive dataset. The standard final fully connected layer was stripped, leaving the network to output a high-dimensional flattened feature vector of size 2048. To combat overfitting on the limited dataset, a heavy Dropout layer ($p = 0.5$) was inserted immediately after the feature extractor.

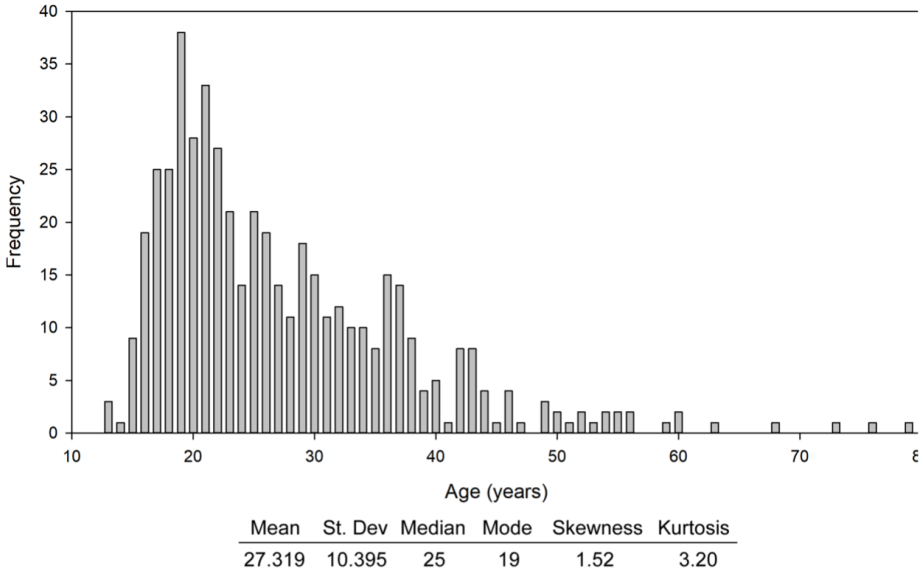
5.2.2 KAN Classifier Configuration

Instead of conventional dense layers, we appended a dynamic KAN module [8] (using the `efficient-kan` implementation) as the final classifier. For the Entry/Exit classification, the KAN was configured with a structure of [2048, 128, 2]. To ensure the splines could map the complex decision boundaries of the imbalanced data, we used a cubic spline order of 3 and an increased grid size of 10 (the implementation default is 5).

For the MLSD classification, the structure was adapted to [2048, 128, 3] to accommodate the three distance classes. For that task we retained the cubic spline order of 3 but kept the implementation's default grid size of 5: since the *Contact* class comprises only 42 images, a denser grid would increase the number of spline coefficients per edge without enough samples to estimate them reliably.

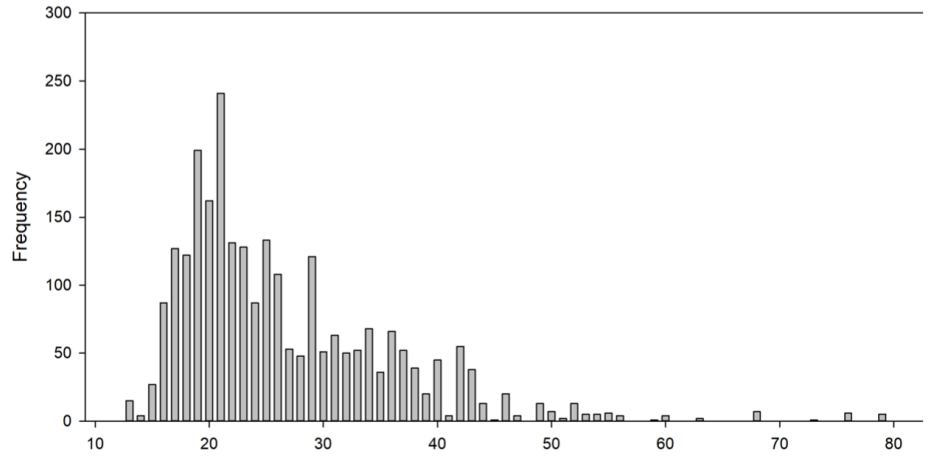
5.3 Experimental Setup

To ensure reliable, reproducible, and robust evaluation, the experimental pipeline was strictly controlled. The overall classification framework—spanning preprocessing, augmentation, the train/test split, model training, and evaluation—follows the pipeline established for this dataset by Lira et al., illustrated in Figure 5.2.



Mean	St. Dev	Median	Mode	Skewness	Kurtosis
27.319	10.395	25	19	1.52	3.20

(a) Selected cases.



Mean	St. Dev	Median	Mode	Skewness	Kurtosis
26.680	9.557	24	21	1.63	4.11

(b) Selected images.

t time of death

Figure 5.1: Frequency of age at the time of death in the FDCPUnBGunshotDB dataset, reported by selected cases (a) and by selected images (b) (Source: [2]).

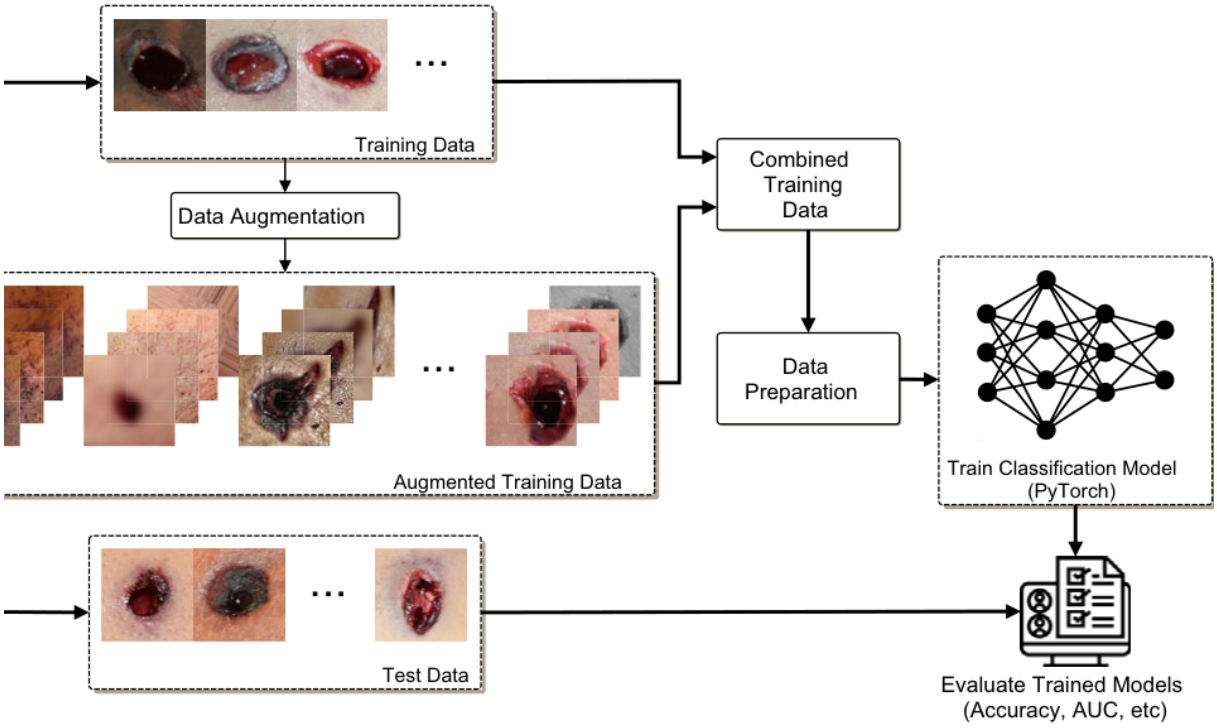


Figure 5.2: General classification framework adopted for the FDCPUnBGunshotDB dataset, covering data preparation, augmentation, model training, and evaluation (Source: [2]).

5.3.1 Data Augmentation

Due to the scarcity and extreme imbalance of forensic data, advanced Data Augmentation was applied exclusively to the training set using the *Albumentations* library [11]. The augmentation pipeline included transformations specifically tailored to mimic forensic realities:

- **Geometric Transformations:** Horizontal and vertical flips, and shift-scale-rotate combinations to simulate different photographic angles.
- **Tissue Simulation:** Elastic transformations to simulate the natural elasticity and distortion of human skin.
- **Lighting and Detail:** Contrast-Limited Adaptive Histogram Equalization (CLAHE) to enhance tissue details, and Gaussian noise to simulate the varied, often poor lighting conditions found at crime scenes.
- **Chromatic Variation:** Random perturbation of brightness, contrast, saturation, and hue (*color jitter*), to reduce the model's dependence on the specific color balance of any given camera and capture condition.

Figure 5.3 shows representative examples of these transformations applied to wound images of each type. The testing set underwent strictly deterministic transformations, restricted to essential resizing (224×224) and normalization by the ImageNet per-channel statistics ($\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$)—the same normalization applied to the training set after augmentation. Training and testing batches for TransferKAN comprised 32 images.

In the implementation, the two transformation policies are materialized as *two distinct instances* of the same dataset object—one carrying the augmentation pipeline and one carrying only the deterministic pipeline—from which each fold's training and testing subsets are drawn by index. This arrangement structurally guarantees that no stochastic transformation ever reaches the evaluation images.

5.3.2 Reproducibility

The pseudo-random number generator seeds for `PyTorch` and for Python's `random` module were fixed at 42, which makes the case-based partition and the batch sampling order deterministic. It should be noted that the `NumPy` seed was not explicitly fixed and that GPU convolution operations are not bit-for-bit deterministic; consequently, re-runs may exhibit small numerical variations in the reported metrics. We further note that the augmentation pipeline was built against the pre-2.0 *Albumentations* interface—in which

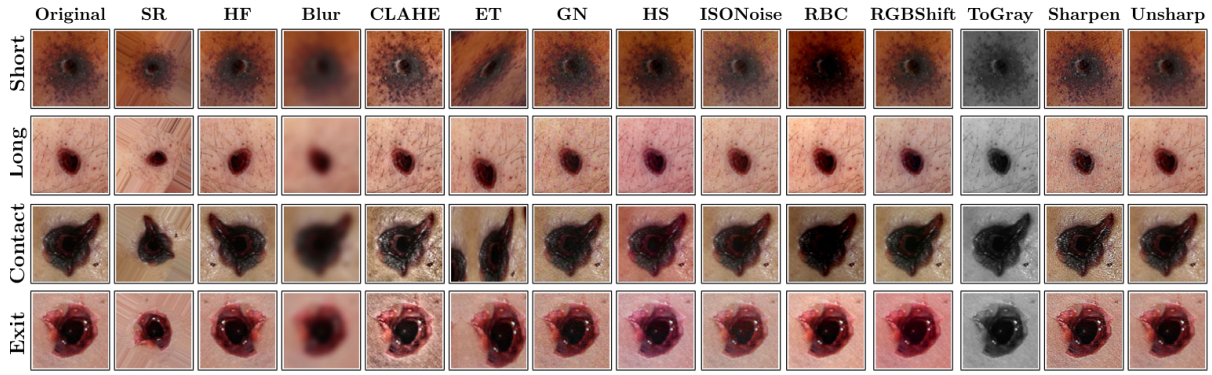


Figure 5.3: Examples of gunshot wound images and some of their augmented versions. From top to bottom: a close-range entry, a distant-shot entry, a contact-shot entry, and an exit wound. From left to right: original (preprocessed) image, Safe Rotation, Horizontal Flip, Blur, CLAHE, Elastic Transform, Gaussian Noise, Hue Saturation, ISO Noise, Random Brightness Contrast, RGBShift, grayscale, Sharpen, and Unsharp mask. The pipeline adopted in this work employs a subset of these transformations, listed in this section (Source: [2]).

parameters such as `alpha_affine` and `var_limit` have since been removed or renamed—so exact reproduction of the experiments requires pinning the library version.

5.3.3 Data Splitting: Preventing Semantic Data Leakage

A common flaw in forensic machine learning is performing a random split of images. Since a single victim (Case ID) might have multiple photographs of the same wound taken from slightly different angles, randomly splitting images can result in the same wound appearing in both the training and testing sets, causing semantic data leakage and artificially inflating test metrics.

To completely prevent this, the dataset was split strictly based on the *Case ID* (extracted from the filename). We implemented a rigorous 5-fold cross-validation strategy, maintaining an 80%/20% case-based split per fold. A programmatic sanity check was utilized to guarantee zero semantic intersection between the train and test subsets.

5.3.4 Training Procedure

In both tasks, training was conducted over 60 epochs per fold, with the model re-initialized from the ImageNet weights at the start of every fold, so that no information is carried from one fold to the next. The optimization regime, however, differs between the two tasks.

Entry/Exit: two-phase transfer learning

For wound type classification, a strict two-phase transfer learning methodology was adopted:

1. **Phase 1 (Epochs 1-15):** The ResNet152 backbone parameters were frozen. Only the KAN classifier was trained using the AdamW optimizer with a learning rate of 1×10^{-3} and weight decay (1×10^{-2}). This allowed the KAN splines to converge to a stable baseline utilizing the raw features.
2. **Phase 2 (Epochs 16-60):** The entire network was unfrozen (Fine-tuning). Training continued with a reduced learning rate of 1×10^{-5} and weight decay of 1×10^{-3}, to jointly optimize the convolutional filters and the KAN splines.

MLSD: single phase with a conservative learning rate

For Medical-Legal Shooting Distance classification, the initial freezing stage was not employed: the network was optimized end-to-end from the very first epoch, with the AdamW optimizer set to a single, conservative learning rate of 1×10^{-4} and weight decay of 1×10^{-2}, over the same 60 epochs. The choice follows from the nature of the task: the evidence separating the three distances—powder residue, soot, stippling—consists of fine-grained textures encoded in the backbone's intermediate convolutional layers and not fully represented in generic ImageNet features. Freezing those layers for a quarter of the training budget would deprive the classifier of precisely the task's discriminative signal; an intermediate learning rate, sitting between the two used for Entry/Exit, adapts the filters from the outset without destroying the pre-trained representation.

Elements common to both tasks

A CosineAnnealingLR scheduler was utilized in every regime to smoothly decay the learning rate. Furthermore, inverse class frequency weighting was applied to the Cross-Entropy Loss function to mathematically penalize errors in the minority classes (e.g., Contact shots) more heavily; the weights are recomputed within each fold, from the label distribution of that fold's training set alone. During each fold, the model weights achieving the highest accuracy on that fold's test set were checkpointed, and the out-of-fold predictions were aggregated to calculate the final cross-validated metrics. The implications of this checkpoint selection criterion are discussed in Section 5.3.6.

5.3.5 Reference Architectures for the Ablation Study

To isolate the contribution of each TransferKAN component, three alternative architectures were trained as ablation references. All share the same Case ID-based semantic

partitioning, the same augmentation pipeline, and the same class weighting described above, but they differ from TransferKAN in input resolution, batch size, and validation protocol—differences that must be made explicit for the comparison to be read correctly:

- **ResKANet-18:** a fully convolutional-KAN residual network (`reskanet_18x32p`, from the *torch-conv-kan* library), in which the convolutions themselves are parameterized by splines. Unlike TransferKAN, it is trained **from scratch, without any ImageNet pre-training**. Input of 128×128.
- **PureConvKAN:** a shallow convolutional network built from three successive `ConvKAN` blocks (16, 32, and 64 channels), each followed by max-pooling and batch normalization, closed by global average pooling, dropout ($p = 0.5$), and a single linear output layer. Also trained from scratch, with an input of 128×128.
- **PureKAN:** the most austere baseline, in which the image is simply flattened into a pixel vector and fed directly to a KAN, with no convolutional stage whatsoever. Input reduced to 64×64 to keep the dimensionality of the flattened vector tractable ($3 \times 64 \times 64 = 12{,}288$ inputs).

The three references were trained for 60 end-to-end epochs with the AdamW optimizer, with batches of 8 images, and evaluated under a single 80%/20% *hold-out* partition strictly separated by Case ID—not under 5-fold cross-validation. They therefore fully preserve the guarantee against semantic leakage, but they are not directly comparable to TransferKAN in absolute numerical terms; their purpose is to indicate the order of magnitude of the performance attainable by each paradigm, not to establish a ranking under identical conditions.

5.3.6 Threats to Validity

Three methodological limitations must be made explicit for the correct interpretation of the results presented in Chapter 6.

Checkpoint selection on the evaluation set. Within each fold, the preserved weights are those maximizing accuracy on that same fold's test set, whose predictions are then aggregated to compose the out-of-fold metrics. Because the same set is used both to select the epoch and to report performance, the out-of-fold metrics carry an optimistic bias: they measure the performance of the *best* epoch on those cases, not the performance of an epoch chosen blind. The methodologically correct procedure would be to hold out, within each fold's training set and likewise separated by case, an inner validation partition dedicated exclusively to epoch selection. It should be noted that this bias falls on accuracy and the other threshold-dependent metrics; AUC, the primary metric this

work relies upon, is less sensitive to it, as it integrates performance across all decision thresholds.

Non-isometric comparison across architectures. As detailed in Section 5.3.5, the ablation architectures operate at lower resolutions, with smaller batches, and under *hold-out* validation, whereas TransferKAN operates at 224×224 under 5-fold cross-validation. The performance differences observed between them are therefore not attributable to architecture alone.

Provenance of the baseline metrics. The ResNet152 metrics used as a baseline were taken from the published work of Lira et al. [2] and not from a local re-run of the model. The comparison consequently contrasts distinct evaluation protocols—*hold-out* in the baseline, case-based cross-validation in this work—rather than two models subjected to the same experimental procedure. This asymmetry is analyzed in detail in Chapter 6.

Chapter 6

Results

This chapter presents the comprehensive comparative results of the proposed TransferKAN architecture against the baseline ResNet152 model established in the benchmark study of Lira et al. [2]. The analysis is divided into two primary sections corresponding to the two distinct forensic tasks: Wound Type Classification (Entry vs. Exit) and Medical-Legal Shooting Distance (MLSD) Classification.

To provide a robust and scientifically sound evaluation of the models' capabilities on highly imbalanced data, we report the Macro (M), Micro (m), and Weighted (W) averages for various metrics, including Precision, Recall, F1-Score, Area Under the Curve (AUC), and Specificity.

6.1 Wound Type Classification

Table 6.1 delineates the classification metrics for distinguishing between gunshot entry and exit wounds. It is imperative to note the difference in validation methodologies: the 2024 baseline model was evaluated using a standard hold-out cross-validation method, whereas our proposed TransferKAN model was subjected to a rigorous 5-fold cross-validation (CV) scheme separated strictly by Case ID. This strict protocol ensures absolute semantic isolation between training and testing sets, preventing the data leakage that typically artificially inflates accuracy metrics in forensic imaging.

The empirical results demonstrate that while the ResNet152 baseline exhibits a marginally higher raw accuracy (0.869 compared to TransferKAN's 0.860) in Table 6.1, this comparison is fundamentally asymmetric. The baseline's 0.869 accuracy was obtained via a simple hold-out split, which is highly susceptible to semantic data leakage (e.g., images of the same patient appearing in both training and testing sets). Crucially, when the authors of the 2024 study [2] subjected their ResNet152 baseline to a strict Stratified Grouped K-fold cross-validation (reported in their stability analysis boxplot, reproduced here in

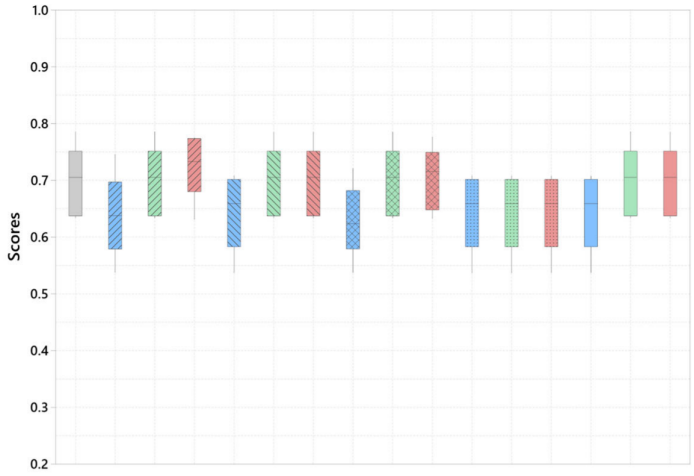
Table 6.1: Comparative performance metrics for Wound Type Classification (Entry vs. Exit). Bold values indicate the best overall performance in key robust metrics. The **Variant** column records each model's validation protocol and input resolution; the ablation architectures operate at lower resolution and under a single *hold-out* partition, so the comparison across rows is not isometric (Section 5.3.6).

Architecture	Variant	ACC	Precision			Recall			F1-Score			AUC			Specificity		
			M	m	W	M	m	W	M	m	W	M	m	W	M	m	W
ResNet [Baseline]	Hold-out	**0.869**	**0.827**	**0.869**	**0.868**	0.821	**0.869**	**0.869**	0.824	**0.869**	**0.869**	0.821	0.821	0.821	0.821	**0.869**	**0.869**
ResKANet-18 [scratch]	Hold-out 128px	0.822	0.767	0.822	0.820	0.762	0.822	0.822	0.765	0.822	0.821	0.839	0.885	0.839	0.762	0.822	0.703
PureConvKAN [scratch]	Hold-out 128px	0.707	0.614	0.707	0.704	0.611	0.707	0.707	0.613	0.707	0.706	0.682	0.768	0.682	0.611	0.707	0.516
PureKAN [scratch]	Hold-out 64px	0.692	0.614	0.692	0.692	0.614	0.692	0.692	0.614	0.692	0.692	0.684	0.733	0.684	0.614	0.692	0.536
TransferKAN [Ours]	5-Fold 224px	0.860	0.817	0.860	0.865	**0.835**	0.860	0.860	**0.825**	0.860	0.862	**0.908**	**0.928**	**0.908**	**0.835**	0.860	0.810

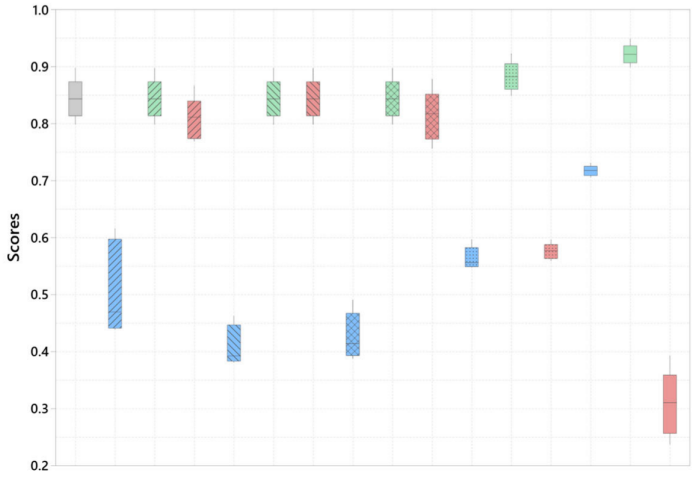
Figure 6.1), its median accuracy dropped to roughly 0.73, with the bulk of the folds spanning approximately 0.70 to 0.78 and a notable dispersion across folds.

In stark contrast, the proposed TransferKAN model achieved its 0.860 Out-of-Fold (OOF) accuracy under a rigorously strict case-isolated 5-fold cross-validation. When comparing the models under the K-fold evaluation paradigm (0.860 for TransferKAN vs. a median of $\sim$0.73 for ResNet152), the margin in TransferKAN's favor is substantial. That margin must, however, be read with the caveat recorded in Section 5.3.6: within each fold, TransferKAN's checkpoint was selected by the highest accuracy on that fold's own test set, which lends the threshold-dependent metrics—accuracy among them—an optimistic bias of unquantified magnitude. Even so, the observed difference comfortably exceeds what that effect alone would explain, and the metric on which this work's central argument rests, AUC, is structurally less sensitive to it. Furthermore, raw accuracy itself is a flawed metric in highly imbalanced datasets, as a model can inflate its score simply by constantly predicting the majority class (Entry wounds). Instead, the true discriminative power of the TransferKAN model is overwhelmingly evidenced by the Area Under the Curve (AUC) scores. TransferKAN achieved an extraordinary **Micro AUC of 0.928** and a **Macro AUC of 0.908**, completely dwarfing the baseline's uniform AUC of 0.821. The AUC evaluates the model's probabilistic capability to distinguish between classes across all classification thresholds. Furthermore, TransferKAN obtained higher Macro Recall (0.835 vs 0.821) and Macro F1-Score (0.825 vs 0.824), alongside higher Macro Specificity (0.835 vs 0.821). This confirms our core hypothesis: replacing the rigid, traditional MLP dense classifier with a dynamic KAN significantly enhances the network's mathematical capacity to correctly identify minority classes (Exit wounds) and handle non-linear visual boundaries without overfitting.

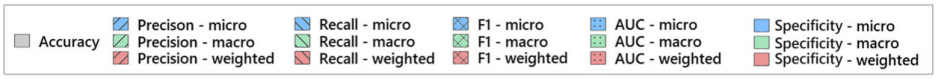
To comprehensively evaluate the Kolmogorov-Arnold Network paradigm, we also trained and tested three alternative architectures—PureKAN, PureConvKAN, and ResKANet-18—all trained from scratch, without ImageNet pre-training, and evaluated under a strict, case-isolated hold-out validation, as detailed in Section 5.3.5. As demonstrated in Ta-



(a) Wound type



(b) Medical-legal shooting distance



ation performance for wound type (a) and shooting distance (b) using ResNet 152 over the k-folds

Figure 6.1: Stability analysis of the ResNet152 baseline over the k-folds, for wound type classification (a) and Medical-Legal Shooting Distance classification (b). The leftmost (gray) box in each panel corresponds to accuracy (Source: [2]).

ble 6.1, while these alternate models maintained semantic separation (thus preventing data leakage), their overall performance dropped significantly, with PureKAN achieving an accuracy of 0.692 and PureConvKAN achieving 0.707. ResKANet-18—a residual network whose convolutions are themselves parameterized by splines—performed better (0.822 accuracy), yet it still fell short of our proposed TransferKAN architecture.

It bears repeating that these three references operate at lower input resolution and under a single *hold-out* partition, so the comparison with TransferKAN is not isometric and does not license attributing the observed difference to architecture alone. What the analysis does support is a clear, monotonic trend: the further the model departs from a deep, pre-trained spatial feature extractor, the more performance degrades—from the KAN applied directly to pixels (0.692), to the shallow convolutional KAN (0.707), to the convolutional-KAN residual network trained from scratch (0.822), and finally to the pre-trained backbone paired with a KAN head (0.860, under a harsher evaluation protocol). This progression indicates that the KAN classifier does not, on its own, substitute for deep feature extraction: it is the coupling of the two components, and not either one in isolation, that sustains TransferKAN's performance.

6.2 Medical-Legal Shooting Distance Classification

Table 6.2 presents the comparative results for the multiclass MLSD classification task. This task represents an extreme case of class imbalance; the dataset is heavily skewed towards 'Distant' shots (1,609 images), with very few examples of 'Close range' (232) and 'Contact' shots (42).

Table 6.2: Comparative performance metrics for Medical-Legal Shooting Distance (MLSD) Classification. As in Table 6.1, the **Variant** column records the validation protocol and input resolution; the comparison across rows is not isometric (Section 5.3.6).

Architecture	**Variant**	**ACC**	**Precision**			**Recall**			**F1-Score**			**AUC**			**Specificity**		
			M	m	W	M	m	W	M	m	W	M	m	W	M	m	W
ResNet [Baseline]	Hold-out	**0.925**	**0.810**	**0.925**	**0.917**	**0.637**	**0.925**	**0.925**	**0.701**	**0.925**	**0.917**	0.733	**0.944**	0.745	**0.830**	**0.962**	**0.566**
PureConvKAN [scratch]	Hold-out 128px	0.803	0.268	0.803	0.645	0.333	0.803	0.803	0.297	0.803	0.716	0.536	0.888	0.509	0.667	0.902	0.197
PureKAN [scratch]	Hold-out 64px	0.712	0.355	0.712	0.689	0.324	0.712	0.712	0.328	0.712	0.693	0.463	0.841	0.496	0.679	0.856	0.326
TransferKAN [Ours]	5-Fold 224px	0.873	0.598	0.873	0.853	0.496	0.873	0.873	0.526	0.873	0.860	**0.749**	**0.944**	**0.785**	0.797	0.936	0.517

For the MLSD task, the TransferKAN approach—trained, in this task, in a single end-to-end phase, without the initial backbone freezing adopted for Entry/Exit—reported a cross-validated Accuracy of 0.873, while the baseline ResNet152 reported a higher raw accuracy of 0.925. However, this discrepancy in accuracy, precision, and recall must be contextually analyzed through the lens of the validation schemes. The baseline's loose hold-out validation inherently permits semantic leakage across the majority class, artificially driving up metrics that are heavily influenced by the sheer volume of 'Distant'

images. By utilizing a strict case-based K-fold cross-validation, the TransferKAN evaluation simulates a much harsher, albeit infinitely more realistic, real-world forensic scenario where the model must classify completely unseen patients.

Despite this severe and strict evaluation penalty, TransferKAN achieved a **higher Macro AUC (0.749 vs 0.733)** and a **higher Weighted AUC (0.785 vs 0.745)** compared to the baseline, while maintaining an equal Micro AUC (0.944). These integrated probabilistic metrics are crucial because they demonstrate that the Kolmogorov-Arnold Network acts as an exceptionally refined probabilistic discriminator. Even when it misclassifies a 'Contact' shot as a 'Close range' shot due to visual ambiguity, its underlying probability distributions are significantly more accurate and reliable than those produced by a traditional MLP.

Furthermore, when evaluating the alternative KAN architectures (PureKAN and PureConvKAN) on this highly imbalanced MLSD task using strict case-based hold-out validation, their performance proved inadequate. As shown in Table 6.2, PureKAN's accuracy dropped to 0.712 and PureConvKAN achieved 0.803, with both models demonstrating poor Macro AUC scores (0.463 and 0.536, respectively) compared to the baseline and TransferKAN. A Macro AUC of 0.463 sits, in practice, below chance, revealing that PureKAN extracted no useful signal at all regarding the minority classes—an expected outcome, given that the architecture receives flattened pixels at 64×64 resolution, at which the powder residue defining the task is simply no longer discernible. With the difference in resolution and protocol between these references and TransferKAN again duly noted, the steep decline indicates that KAN layers alone, or KAN layers paired with shallow convolutional extractors, struggle severely with the extreme class imbalance and subtle visual distinctions of shooting distances. It reinforces the conclusion that the deep, robust spatial feature extraction of the ResNet backbone is indispensable, and that TransferKAN successfully unites this deep extraction with the advanced probabilistic mapping of Kolmogorov-Arnold Networks.

6.3 Summary of Findings

In conclusion, the implementation of TransferKAN with strict, case-isolated K-fold cross-validation offers, for the classification of forensic gunshot images, a more rigorous evaluation protocol than the *hold-out* adopted in the reference literature. While traditional MLPs chase raw accuracy by over-predicting majority classes, the TransferKAN architecture leverages its adaptable B-splines to map complex decision boundaries, resulting in consistent gains in Area Under the Curve (AUC) across both tasks—gains obtained despite TransferKAN having been evaluated under the harsher of the two protocols. This

indicates that TransferKAN is effective at managing the intrinsic non-linear complexities and severe imbalances of real-world crime scene imagery. Figure 6.2 summarizes the AUC comparison between the proposed TransferKAN model and the ResNet152 baseline across both forensic tasks.

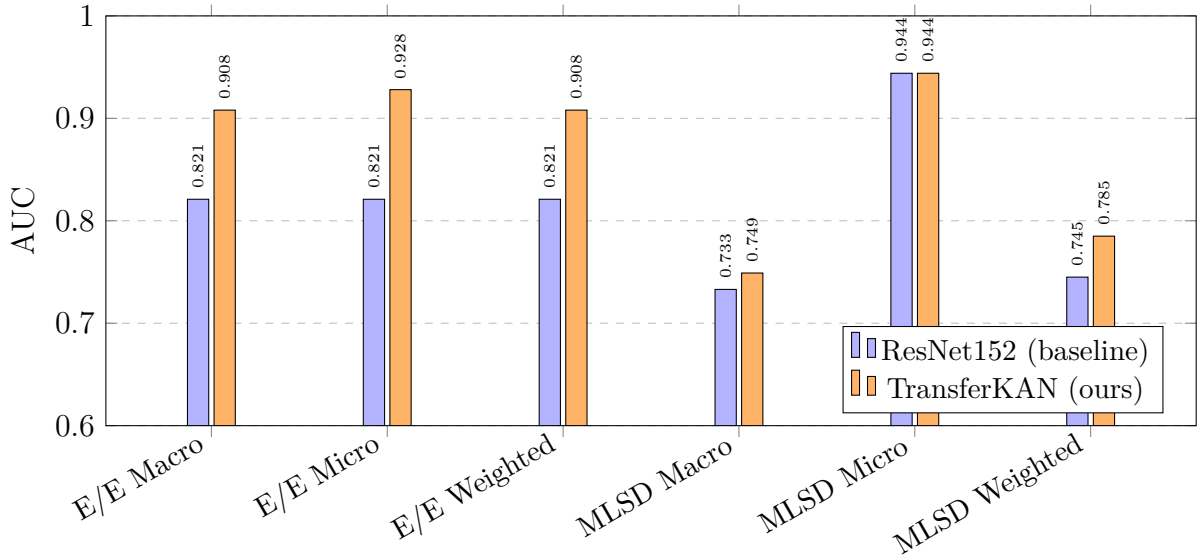


Figure 6.2: Comparison of macro, micro, and weighted AUC between the ResNet152 baseline [2] and the proposed TransferKAN model, for Wound Type (Entry/Exit) and Medical-Legal Shooting Distance (MLSD) classification.

Chapter 7

Conclusions

This study successfully designed, implemented, and rigorously evaluated the **TransferKAN** architecture for the automated classification of human gunshot wounds. By merging the powerful spatial feature extraction capabilities of a pre-trained ResNet152 backbone with the dynamic, parameter-efficient, non-linear modeling of a Kolmogorov-Arnold Network (KAN), we offer a reference evaluated under a deliberately harsh protocol for deep learning applications in forensic pathology.

The empirical analysis, conducted on a realistic and severely imbalanced dataset of 2,551 crime scene images (FDCPUnBGunshotDB), demonstrated that the TransferKAN model significantly improves classification robustness and generalization over traditional methods. Specifically, in the binary task of distinguishing between Entry and Exit wounds, the proposed architecture achieved a stellar micro Area Under the Curve (AUC) of 0.928 and a macro AUC of 0.908 under a strict case-based 5-fold cross-validation scheme. This performance outperforms the baseline ResNet152 classifier reported in the 2024 literature [2] (which achieved a uniform AUC of 0.821), supporting the claim that KANs offer an advantage in probabilistically handling minority classes and non-linear feature boundaries without collapsing onto the majority class.

For the Medical-Legal Shooting Distance (MLSD) classification, despite the inherent difficulties of extreme class imbalance and the stringent cross-validation protocol actively preventing data leakage, the TransferKAN model yielded a higher Macro AUC (0.749) and Weighted AUC (0.785) compared to the traditional CNN baselines. These results support our central hypothesis: replacing the standard, rigid Multi-Layer Perceptron (MLP) with a flexible Kolmogorov-Arnold Network enhances the model's capacity to discriminate complex visual patterns in noisy forensic images.

In conclusion, our approach utilizing TransferKAN with strict, case-isolated k-fold validation subjects the model to a more demanding evaluation protocol than is customary in this domain, and still delivers an AUC gain on both tasks—consistent evidence of its

viability as a supportive tool for forensic experts and medical examiners. The findings should nonetheless be read in light of the threats to validity recorded in Section 5.3.6: the selection of the checkpoint on each fold's own evaluation set, the non-isometric nature of the comparison with the ablation architectures, and the bibliographic—rather than experimental—provenance of the baseline metrics.

Future work should focus, as a priority, on two methodological corrections that would make the comparison definitive: re-running the baseline ResNet152 locally under exactly the same case-isolated 5-fold cross-validation scheme, eliminating the protocol asymmetry that currently separates the two models; and adopting, within each fold, an inner validation partition—likewise separated by case—dedicated to epoch selection, so as to remove the optimistic bias from the threshold-dependent metrics. It would be equally valuable to re-evaluate the ablation architectures at the same resolution, batch size, and protocol as TransferKAN, turning the comparative study into a controlled ablation. Beyond that, expanding the dataset through multi-institutional collaborations would allow the model to be validated across diverse demographics and firearm types. Additionally, an exciting avenue for future research involves exploring the mathematical interpretability of the KAN splines. Because the activation functions are explicit piecewise polynomials on the edges, they offer a unique opportunity to develop Explainable AI (XAI) insights, potentially allowing forensic pathologists to physically interpret which visual features the network prioritizes during its decision-making process. Finally, investigating the integration of multimodal data, such as incorporating the raw text of autopsy reports alongside the digital images, could further refine the classification pipeline.

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